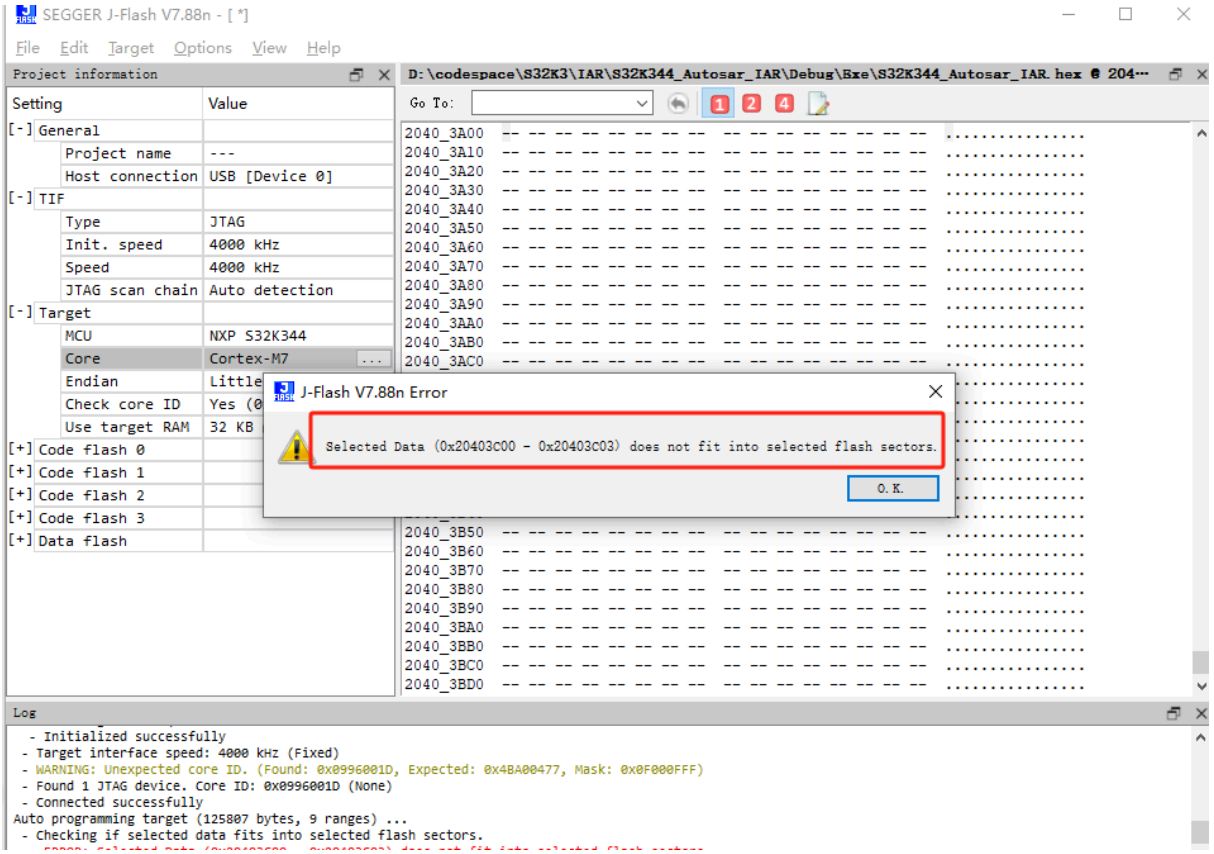
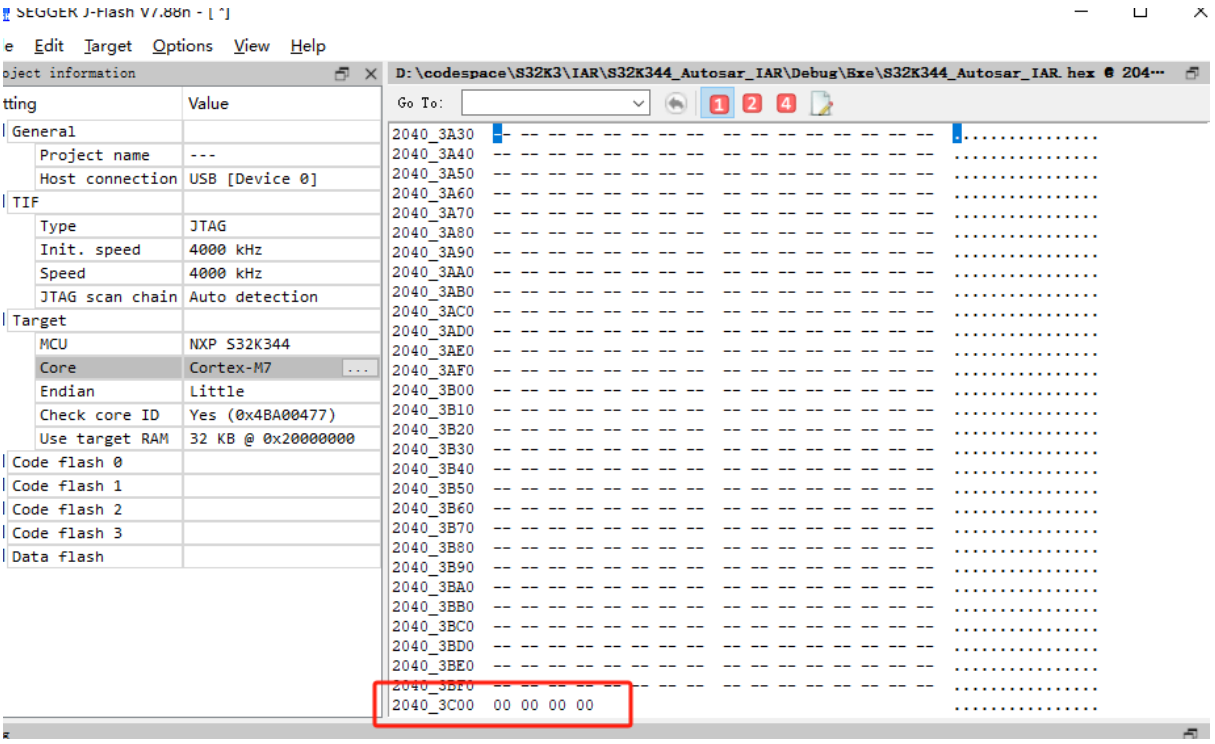
12、加入CAN功能后编译hex有超范围到RAM的问题分析

卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

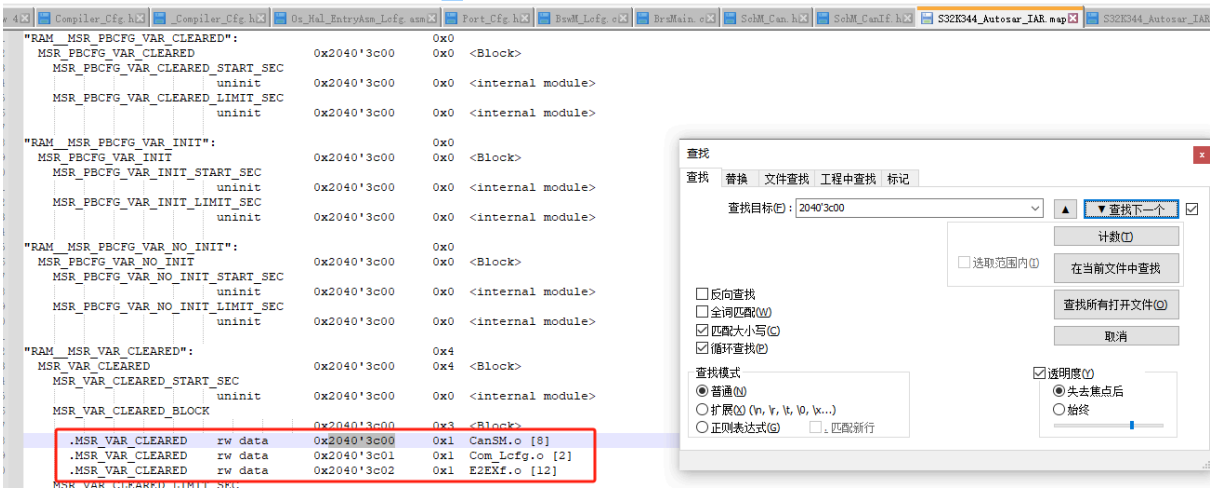
使用Flash烧录时报错，生成的Hex 有4字节是不在Code flash区间内， 可见0x20403C00~0x20403C03是RAM段地址，不应该出现在CodeFlash里面



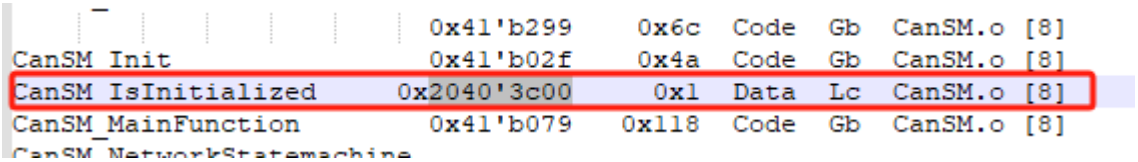


一、倒推分析产看map文件

我们知道，变量都会在map文件里面有映射，我们先去map文件找这四个字节到底是谁用



从map文件可以找到，.o 里面有三个变量用了该ram地址



```
Com_IPduGroupState      0x2040'3cec    0x2   Data   Gb   Com_Lcfg.o [2]
Com_Init                  0x41'750d   0x3c   Code   Gb   Com.o [9]
Com_Initialized         0x2040'3c01    0x1   Data   Gb   Com_Lcfg.o [2]
Com_IpduGroupStart        0x41'73e1   0x90   Code   Gb   Com.o [9]
Com_IpduGroupStartRx      0x41'7db5   0xc2   Code   Gb   Com_IPduGroupHdlr.
```

```
E2EXf_Init_Calls          0x41'b4fd    0x2   Code   Gb   E2EXf_LCfg.o [2]
E2EXf_ModuleInitialized
                        0x2040'3c02    0x1   Data   Lc   E2EXf.o [12]
ESH_ComM_CheckPendingRequests
                          0x41'b4d1   0x2c   Code   Gb   BswM_Callout_Stubs.o [4]
```

他们对应用的段 .MSR_VAR_CLEARED ，所以七有八九是这个段配置的异常导致的，我们以CanSM.o 里面的CanSM_IsInitialized说明

二、找出对应的段名的流程

```
CanSM.c | Com_Lcfg.c | MemMap.h | vLinkGen_Template.icf | BrsMain.c | EcuM_Callout_Stubs.c | EcuM.c | Port.c | ARMStartup_CortexM.c | CtApTestSwc.c | Rte.c | Uart.c | Os_Hal_

CANSM_LOCAL P2CONST( CanSM_ConfigType, CANSM_VAR_NOINIT, CANSM_CONST ) CanSM_ConfigDataPtr;

# define CANSM_STOP_SEC_VAR_NOINIT_UNSPECIFIED
# include "MemMap.h"  /* PRQA S 5087 */ /* MD_MSR_MemMap */
#endif /* ( CANSM_USE_INIT_POINTER == STD_ON ) */

#define CANSM_START_SEC_VAR_ZERO_INIT_8BIT
#include "MemMap.h"  /* PRQA S 5087 */ /* MD_MSR_MemMap */
/*! Used values are CANSM_INITFUNCTION_NOT_CALLED, CANSM_PRE_INITIALIZED, CANSM_INITFUNCTION_RUNNING, CANSM_INITFUNCTION_FINISHED */
CANSM_LOCAL VAR( uint8, CANSM_VAR_ZERO_INIT ) CanSM_IsInitialized = CANSM_INITFUNCTION_NOT_CALLED;
/* \trace SPEC-11327, SPEC-47131 */

#define CANSM_STOP_SEC_VAR_ZERO_INIT_8BIT
#include "MemMap.h"  /* PRQA S 5087 */ /* MD_MSR_MemMap */
```

这里会里面使能该SEC_VAR_ZERO_INIT_8BIT的段

这里会把段恢复为默认段

我们到MemMap.h 里面找

```
CanSM.c | Com_Lcfg.c | MemMap.h | vLinkGen_Template.icf | BrsMain.c | EcuM_Callout_Stubs.c | EcuM.c | Port.c | ARMStartup_CortexM.c | CtApTestSw

/* VAR ZERO INIT sections */

#ifdef CANSM_START_SEC_VAR_ZERO_INIT_8BIT
# undef CANSM_START_SEC_VAR_ZERO_INIT_8BIT
# define START_SEC_VAR_ZERO_INIT_8BIT
#endif

#ifdef CANSM_STOP_SEC_VAR_ZERO_INIT_8BIT
# undef CANSM_STOP_SEC_VAR_ZERO_INIT_8BIT
# define STOP_SEC_VAR
#endif
```

追踪，MemMap.h 会把CanSM.c define的宏查找，然后对应把头去掉再定义一遍，如START_SEC_VAR_ZERO_INIT_8BIT， STOP_SEC_VAR，然后再去掉davinci生成的MemMap_Common.h里面

```
# undef VSTDLIB_STOP_SEC_CONST_8BIT                    /* PRQA S 0841 */ /* MD_MSR_Undef */
# define STOP_SEC_CONST                                /* default const stop section */
#endif

/*****************************************************************************************************************
 *  VSTDLIB END
 *****************************************************************************************************************/


/* PRQA L:MEMMAP_0841_TAG */

#ifndef MEMMAP_REMAPPING_ONLY
# include "MemMap_Common.h"

/* MEMMAP_ERROR is
   - still defined if MemMap_Common.h did not contain a matching section
   - undefined, if MemMap_Common.h handled the section define. */
# ifdef MEMMAP_ERROR
/* contains all configured MemMap Sections */
#  include "MemMap_Compatibility.h" /* generated by module MemMap */
# endif

# ifdef MEMMAP_ERROR
//#  error No MemMap section found in MemMap_Common.h and MemMap_Compatibility.h. Check your section define for validity.
# endif
#endif
```

MemMap_Common.h × | MemMap.h | vLinkGen_Template.icf | BrsMain.c | EcuM_Callout_Stubs.c | EcuM.c | Port.c | ARMStartup_CortexM.c | CtApTestSwc.c | Rte.c | Uart.c | Os_Hal_Core.c

```
# undef STOP_SEC_VAR_NO_INIT_UNSPECIFIED /* PRQA S 0841 */ /* MD_MSR_Undef */
# undef MEMMAP_ERROR /* PRQA S 0841 */ /* MD_MSR_Undef */

/* ---------------------------------------------------------------------------- */

#elif defined START_SEC_VAR_ZERO_INIT_8BIT
# ifdef MEMMAP_SECTION_OPEN
#  error Using START_SEC_VAR_ZERO_INIT_8BIT is not possible as a memory section has already been opened. Nesting is not supported.
# endif
# define MEMMAP_SECTION_OPEN
# define VAR_OPEN

/* Pragmas generated automatically. Related MemMapAddressingMode: /ActiveEcuC/MemMap/MSR_VAR_CLEARED/MSR_VAR_CLEARED */
# pragma default_variable_attributes = @ ".MSR_VAR_CLEARED" /* PRQA S 0299, 3116 */ /* MD_Os_Dir1.1_0299_Pragma, MD_MSR_Pragma */

# undef START_SEC_VAR_ZERO_INIT_8BIT /* PRQA S 0841 */ /* MD_MSR_Undef */
# undef MEMMAP_ERROR /* PRQA S 0841 */ /* MD_MSR_Undef */

#elif defined STOP_SEC_VAR_ZERO_INIT_8BIT
# ifndef MEMMAP_SECTION_OPEN
#  error Using STOP_SEC_VAR_ZERO_INIT_8BIT is not possible as no memory section has been opened.
# endif
# undef MEMMAP_SECTION_OPEN /* PRQA S 0841 */ /* MD_MSR_Undef */
# ifndef VAR_OPEN
#  error Using STOP_SEC_VAR_ZERO_INIT_8BIT is not possible as the corresponding memory section has not been opened.
# endif
# undef VAR_OPEN /* PRQA S 0841 */ /* MD_MSR_Undef */

/* Pragmas generated automatically. Related MemMapAddressingMode: /ActiveEcuC/MemMap/MSR_VAR_CLEARED/MSR_VAR_CLEARED */
# pragma default_variable_attributes = /* PRQA S 3116 */ /* MD_MSR_Pragma */

# undef STOP_SEC_VAR_ZERO_INIT_8BIT /* PRQA S 0841 */ /* MD_MSR_Undef */
# undef MEMMAP_ERROR /* PRQA S 0841 */ /* MD_MSR_Undef */

/* ---------------------------------------------------------------------------- */
```

start

定义变量时，相当于变量会定义到这个段

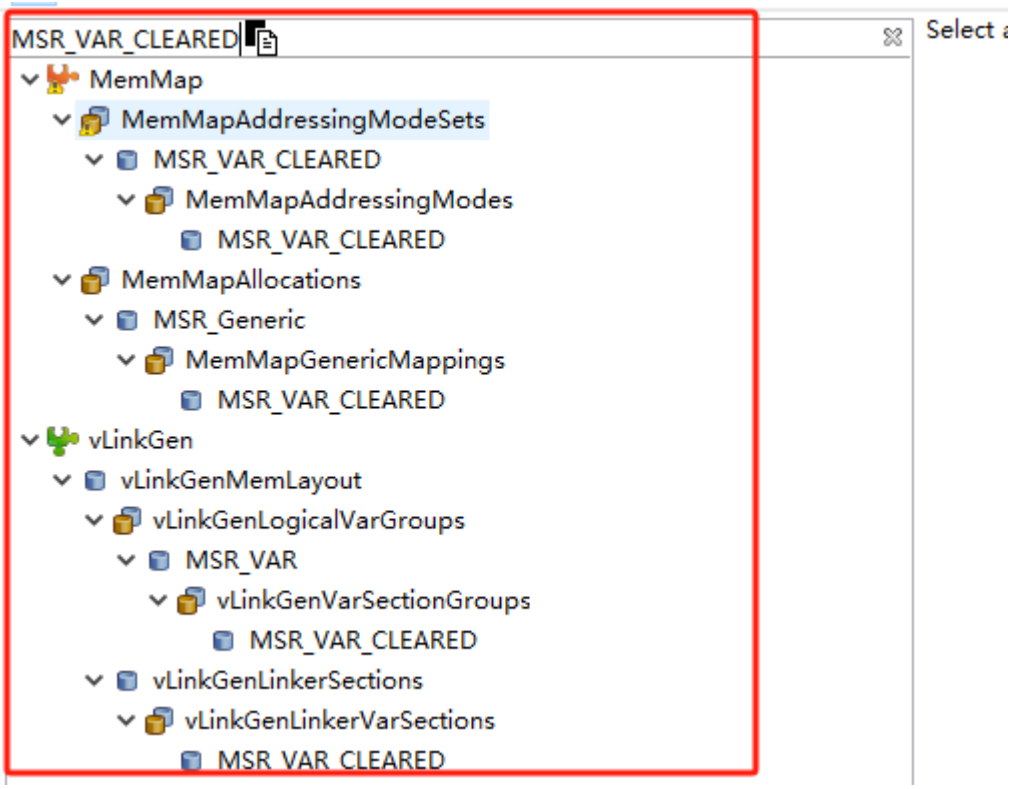
stop

定义变量结束后，把变量恢复为默认段，一般是.data

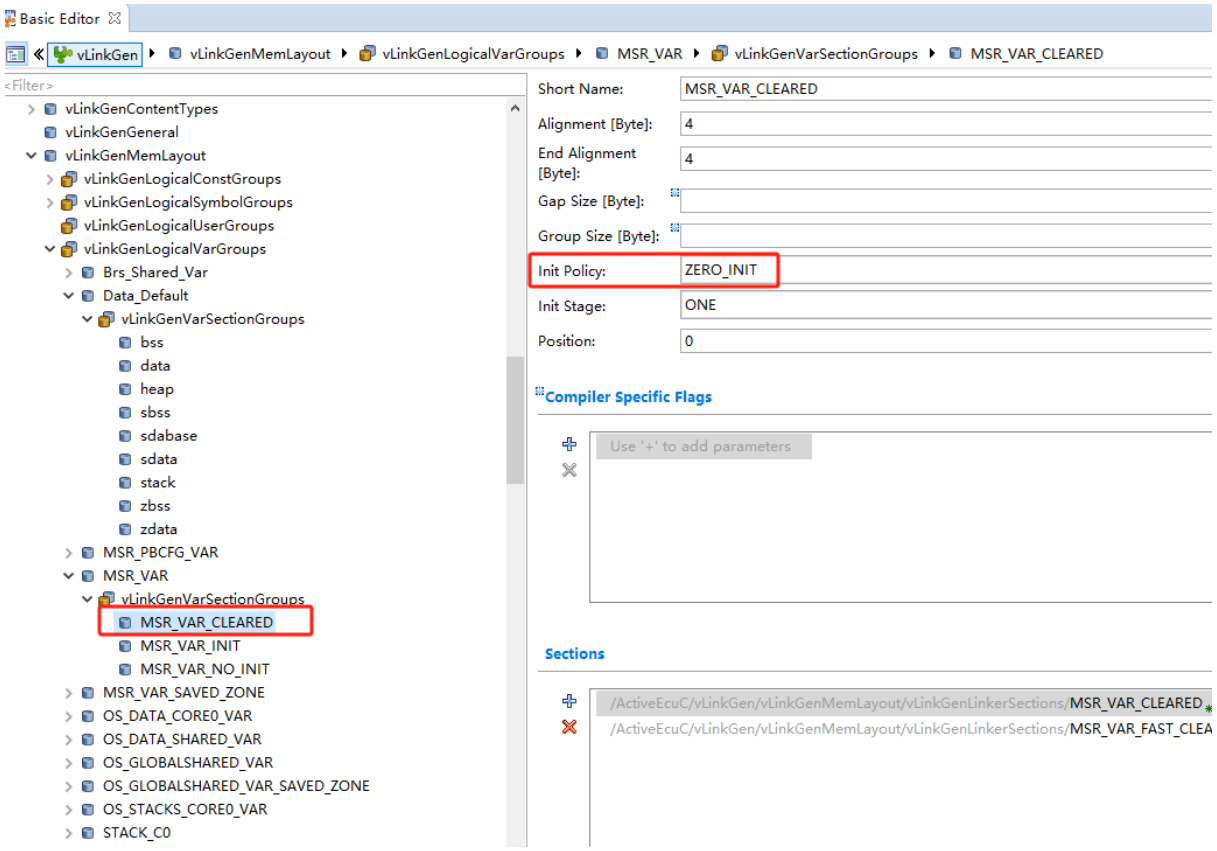
可见 .MSR_VAR_CLEARED 这个段最终是在MemMap_Common.h 里面#pragma 使能了，所以CanSM.o 里面的CanSM_IsInitialized就会分到这个段里面，在stop的定义里面会#pragma会默认段

三、解决方法

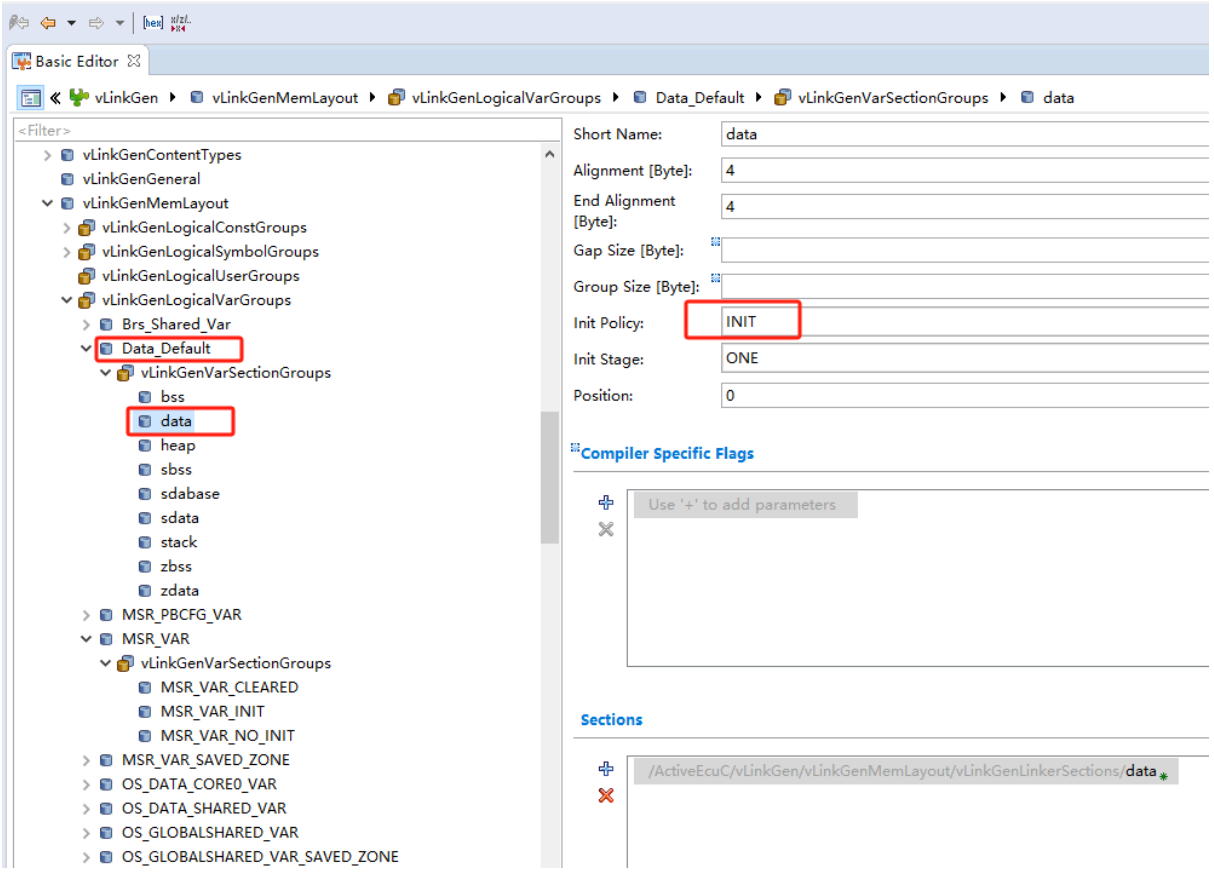
打开davinci, 查看MSR_VAR_CLEARED， 全局搜索MSR_VAR_CLEARED



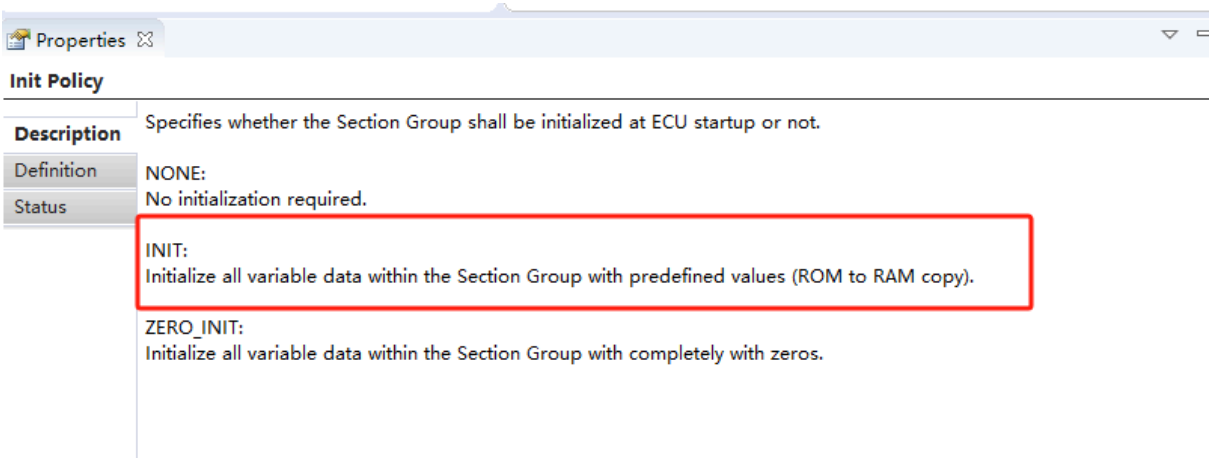
我们对比后发现，因为上面的定义都是全局初始化变量，相当于 .data段的效果



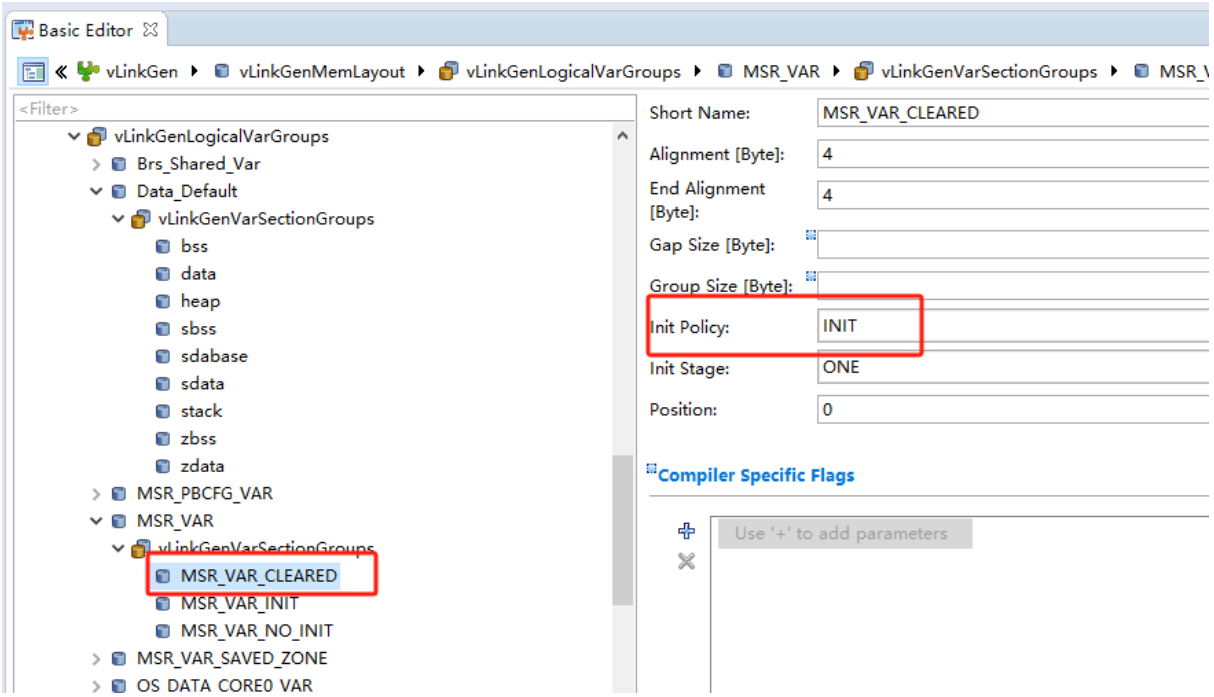
我们对比下.data段的配置



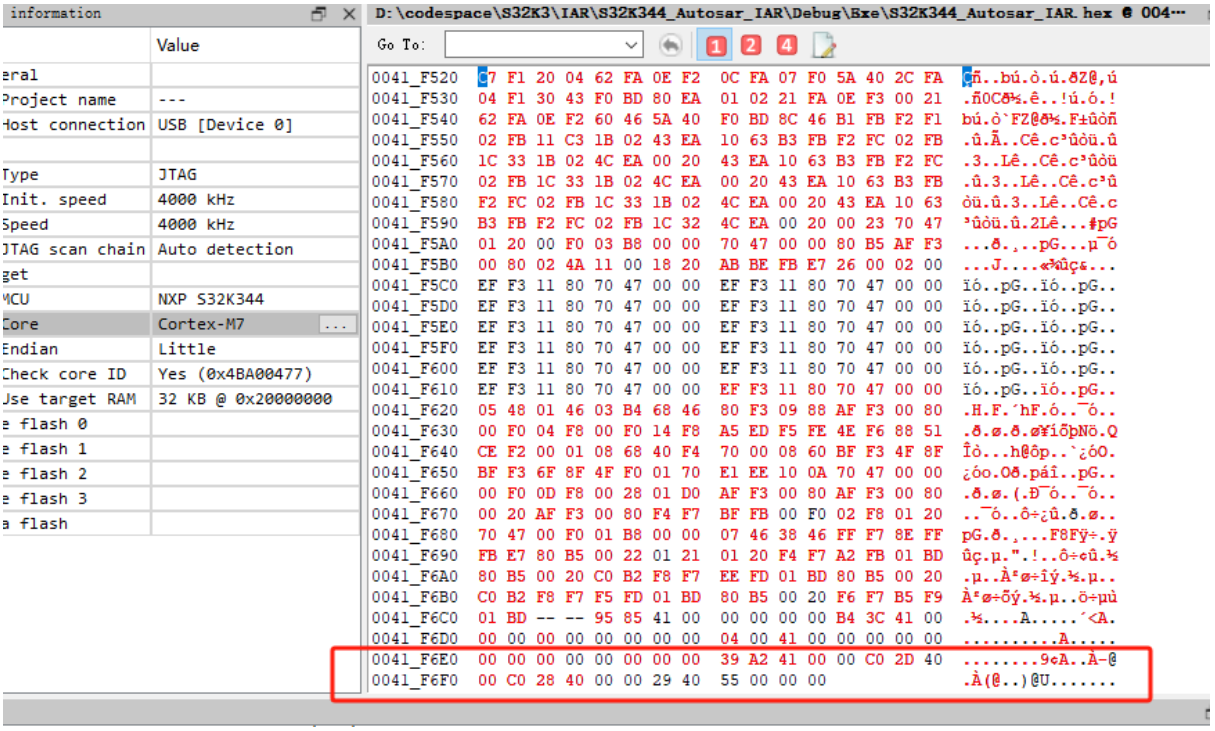
差别就是在 Init Policy 不一样，



其实我们就是想要像.data段的效果， 全局初始化的变量在ram有一份，在rom有一份，rom的那份就是我们默认初始化的值

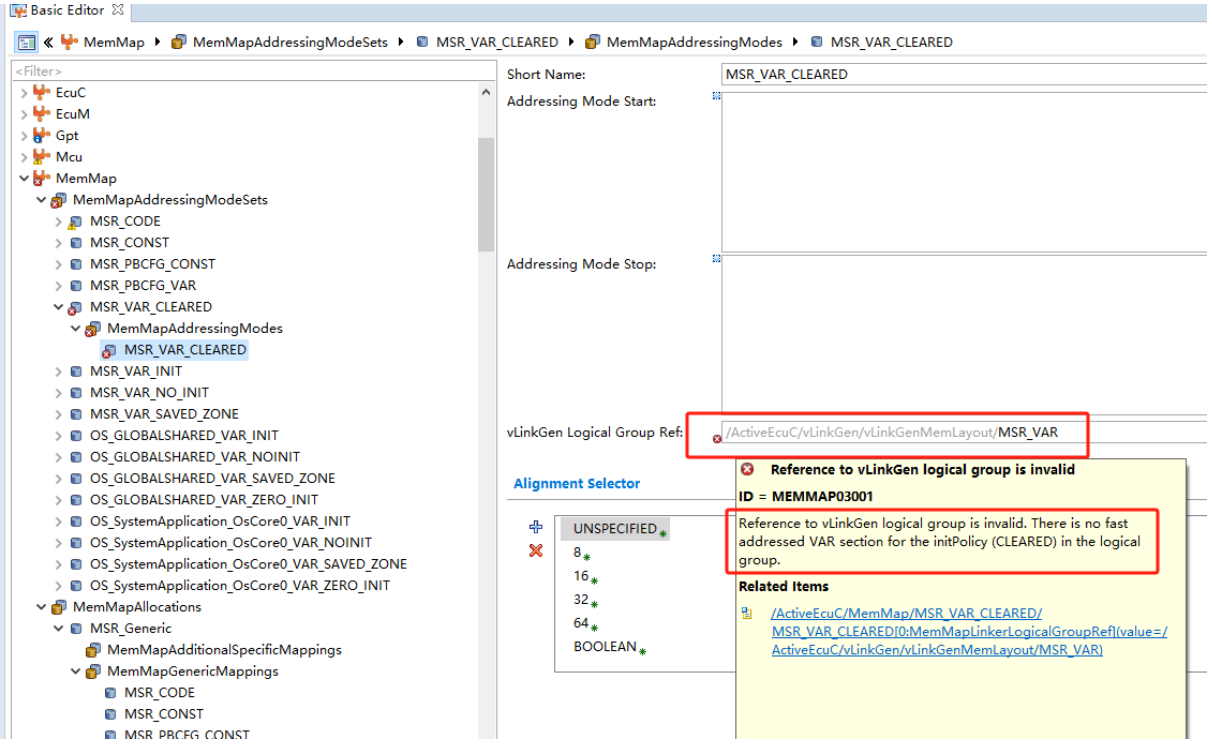


我们修改MSR_VAR_CLEARD 的 Init Policy和data段的一样为 INIT, 然后重新编译vLinkGen，最后重新编译代码出来的hex 就不会超地址范围了。Hex最后的地址都是在CodeFlash范围内了

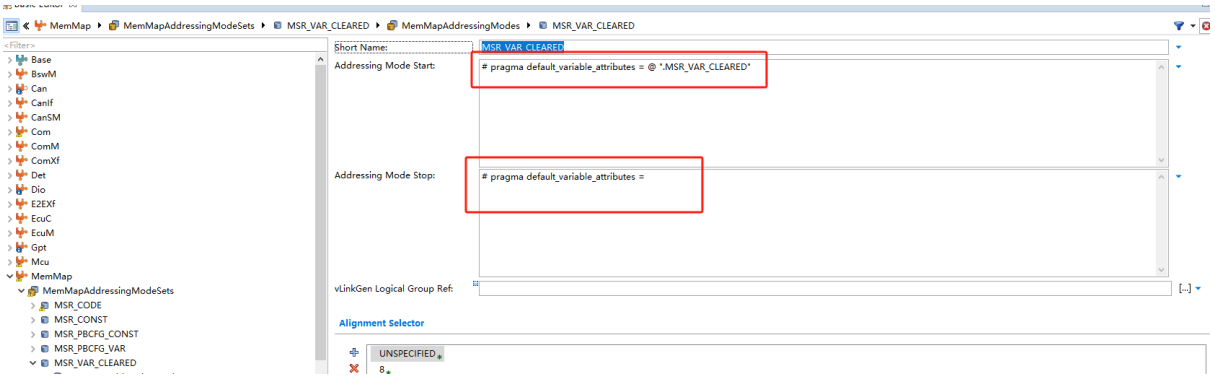


四、最终解决方式

三的方式会导致MemMap模块报错，如下



同时修改MemMap的



然后重新生成MemMap即可